

Study Project Report in the Module
MHSE09 “Study Project”
Chair of Process Engineering in Hydrosystems

Study Project topic: Hydraulic Design of Sedimentation Basin using ANSYS - Fluent

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Study Project Report

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Acknowledgement

Each author contributed to different parts of this work. Fashli Adli Wal Ikhsan was responsible for the Introduction, Literature Review, Theoretical Background, the laboratory experimental setup, Image Processing in FIJI, CFD model configuration, Mesh Independence Study, and compiling the References. Arjun Parajuli was responsible for the experimental and numerical Results and Discussion, including the RTD analysis, model validation, velocity and pathline evaluations, operational sensitivity analysis, as well as the Conclusions and Engineering Implications.

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Abstract

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Residence Time Distribution (RTD) analysis was employed to quantify key hydraulic indicators across unbaffled and baffled geometric configurations. The CFD model was successfully validated against experimental datasets, achieving a minimal relative deviation of $< 6\%$ in mean residence time ($t = 2883 \text{ s}$ experimental vs. 3045 s numerical) and $< 12\%$ in peak exit time. Experimental findings demonstrate that installing an internal deflector baffle mitigates short-circuiting, reducing the stagnant dead zone volume by 77.4% (from 25.64% down to 5.79%) and increasing active volumetric efficiency from 74.36% to 94.21% . Velocity evaluations revealed a fundamental design trade-off: higher inlet momentum eliminates dead space but dramatically elevates hydraulic dispersion (Morrill Index increasing from 2.57 to 7.71). Consequently, internal geometric modification via baffle positioning provides a robust, effective mechanism for optimizing basin hydraulics without compromising the clarification.

Keywords: Sedimentation Basin; Computational Fluid Dynamics (CFD); ANSYS Fluent; Residence Time Distribution (RTD); Baffle Optimization; Hydraulic Efficiency.

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v_s	Particle settling velocity	m/s
V	Total basin volume	m ³
V_d	Dead zone volume	m ³
V_e	Active / effective volume	m ³
μ	Dynamic viscosity	Pa · s
ρ	Fluid density	kg/m ³
σ^2	Hydraulic variance	s ²
σ_θ^2	Dimensionless variance ($\sigma^2/\bar{t}^2$)	—
τ	Theoretical hydraulic retention time (V/Q)	s
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<i>SDG</i>	<i>Sustainable Development Goals</i>
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1 Introduction

Europe Sustainable Development Goals (SDG) are a comprehensive set of legal commitments and targets integrated into EU policy to balance social, economic, and environmental protection, aiming to meet current needs without harming future generations. Clean water and sanitation are one of the 17 SDG that plays a big role. Core aspects in this goal are to ensure safe drinking water, sanitation and hygiene, water quality, water efficiency, water management, and ecosystem protection.

To ensure these core aspects, it is crucial for every water plant, including Drinking Water Plants and Wastewater Treatment Plants (WWTPs) to process the incoming water, so the water product meets the regulations and is safe for the people and the environment. One of the most important aspects that defines the quality of the water is the presence of solid materials. Here, sedimentation basin plays a vital role.

1.1 Problem Statement

In drinking water and wastewater treatment, sedimentation basins serve as a critical physical unit process for the gravitational removal of suspended solids. Historically, the design of these basins has relied heavily on theoretical surface overflow rates based on the assumption of ideal plug-flow conditions. However, physical basins rarely operate under these ideal constraints. Internal hydrodynamics are frequently disrupted by severe short-circuiting, stagnant dead zones, and secondary recirculation loops. These non-ideal flow behaviors significantly reduce the active volumetric efficiency of the basin, impair its gravitational settling capacity, and ultimately degrade the quality of the treated effluent. Addressing these internal flow inefficiencies and maximizing the active settling volume without requiring extensive, costly infrastructural expansions remains a fundamental challenge in environmental engineering.

1.2 Research Gap

Current methodologies for evaluating and optimizing sedimentation basin hydraulics typically rely on either purely empirical Residence Time Distribution (RTD) tests or standalone numerical simulations. While experimental RTD analysis effectively quantifies overall hydraulic retention, it lacks the comprehensive spatial resolution required to visually identify and pinpoint internal flow anomalies or stagnant dead zones.

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Conversely, while Computational Fluid Dynamics (CFD) offers deep hydrodynamic spatial tracking, it demands rigorous physical validation to serve as a reliable digital model. Furthermore, while the general implementation of baffles has been explored, there is a distinct lack of coupled experimental-numerical studies that explicitly quantify the operational trade-offs of adjusting inlet flow conditions. Specifically, current literature does not sufficiently quantify the fundamental design penalty where increasing inlet velocity successfully eliminates dead space but simultaneously triggers severe back-mixing and elevated hydraulic dispersion, necessitating structural rather than purely operational interventions.

1.3 Aims and Purposes

1. Quantify hydraulic parameters using flow field indicators and Residence Time Distribution (RTD).
2. Analyse the impact of geometry on short circuiting, dead zones, and recirculation zones.
3. Analyse the effect of operating condition on the hydraulic parameters.

1.4 Objectives

1. Evaluate and optimize the hydraulic efficiency of a 23.46 L bench-scale sedimentation basin, by combining physical tracer stimulus-response experiments with three-dimensional Computational Fluid Dynamics (CFD) modelling in ANSYS Fluent.
2. Quantify hydraulic parameters via flow field indicators and Residence Time Distribution (RTD).
3. Analyse how physical geometry adjustments impact internal flow dynamics, specifically short-circuiting, dead zones, and recirculation zones.
4. Evaluate the effects of operating conditions, such as variable flow velocities, on overall hydraulic parameters.

1.5 Novelty

1. Couple physical lab-scale tracer experiments, using FIJI image processing to extract mean light intensity as a proxy for concentration, with strict 3D CFD finite-volume simulations to cross-validate hydraulic behaviour.
2. Quantify a fundamental operational trade-off, demonstrating mathematically that while increasing inlet velocity eliminates stagnant dead space (from 40.1% down to ~0.0%), it severely penalizes overall hydraulic efficiency by driving the Morrill Dispersion Index from 2.57 up to 7.71.

1.6 Research Questions

1. How does the installation of an internal deflector baffle alter the physical and numerical Residence Time Distribution (RTD), specifically regarding active volumetric efficiency and early tracer breakthrough?
2. To what degree of accuracy can a single-phase, laminar 3D CFD model replicate the experimental Residence Time Distribution (RTD) curves of a physical dye tracer?
3. How do variations in operational inlet velocity (ranging from 0.10 m/s to 1.00 m/s) dictate the balance between internal flow dispersion and stagnation?

1.7 Hypothesis

1. Modifying the basin's internal geometry via a baffle will intercept high-momentum inlet flow, force fluid downward, and reduce short-circuiting and dead volume compared to an unbaffled configuration.
2. A correctly discretized CFD model using the Finite Volume Method will successfully reproduce physical flow fields and function as an accurate model for evaluating sedimentation basin hydraulics.
3. Relying exclusively on operational flow rate adjustments without geometric modification will fail to yield optimal plug-flow conditions, as increasing kinetic energy triggers back-mixing and elevated dispersion.

1.8 Literature Review

Historically, sedimentation tank design relied primarily on the theoretical surface overflow rate, assuming ideal plug-flow conditions. However, field evaluations and numerical simulations consistently show that full-scale sedimentation basins do not follow ideal flow behaviour, as internal hydrodynamics are disrupted by velocity variations and turbulent shear stresses (Naser et al., 2007).

While experimental Residence Time Distribution (RTD) analysis effectively characterizes overall hydraulic retention based on tracer dispersion, it lacks complete spatial resolution of the internal flow and concentration fields (Nemri et al., 2013). Coupling physical RTD tracer tests with 3D CFD simulations resolves this bottleneck, allowing for deep hydrodynamic analysis that identifies dead zones and tracks velocity profiles natively (Blanco-Aguilera et al., 2020).

The geometry modification using upper and lower baffle arrangements is empirically proven to modify the hydraulics in a sedimentation basin. Al-Mafraji and Al-Mussawy (2021) simulated the physical model of a rectangular sedimentation basin and found that this geometry configuration increased the hydraulic residence time from 1.33 hours (without baffle) to 4 hours (with baffle). It also increases the turbidity removal rate to 25% without any coagulant. They also found that the turbidity removal is more efficient at lower discharge (0.4 L/min) compared to higher discharge (1.0 L/min).

The efficacy of an inlet baffle is strictly dependent on its ability to dissipate the kinetic energy of the influent stream and prevent feedwell backflow. Multi-phase 3D CFD tracking demonstrates that unconventional top-side feed designs lead to significant flow asymmetry. Introducing geometry modifications, such as inward baffles and submerged skirts, successfully dissipates turbulence kinetic energy, creates a uniform flow distribution, and reduces the velocity of the flow exiting the feedwell, leading to a reduction in effluent total suspended solids (TSS) by more than 80% (Das et al., 2015).

Goula et al. (2008) evaluated using CFD the hydraulics and particle removal efficiency in a full-scale sedimentation tank for a drinking water treatment plant. They studied the effect of the extended feed flow control baffle using low particle fraction, so the coagulation and hindered settling effect can be neglected. Result shows that introduction of the baffle increases the kinetic energy dissipation at the inlet and reduces the recirculation zone. This modification also increases the removal efficiency from 90.4% to 98.6% for smooth particles ($<150\text{ }\mu\text{m}$).